

PM10 Pollution Forecasting in Beijing

Guanyuan Station, Beijing, China

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Introduction

This report is prepared for someone who is interested in understanding and forecasting air pollution levels using data from the Guanyuan monitoring station in Beijing, China. Air pollution is a critical environmental and public health issue, particularly in densely populated and industrialized cities such as Beijing. One of the most widely used indicators of air quality is PM10, which refers to particulate matter with a diameter of less than 10 micrometers. These particles are small enough to be inhaled into the respiratory system, where they can cause or worsen conditions such as asthma, bronchitis, and other cardiovascular and respiratory diseases.

Monitoring and forecasting PM10 levels is important because pollution levels can fluctuate significantly from day to day. Accurate predictions allow individuals to take precautionary measures, such as limiting outdoor activities during periods of poor air quality, and can also assist policymakers in implementing timely environmental interventions.

The primary objective of this project is to develop predictive models that can forecast PM10 levels for the following day using historical pollution and weather data. In addition to achieving strong predictive performance, an important goal of this analysis is to ensure that the results are communicated clearly and can be interpreted by a general audience. This allows the findings to be translated into practical, real-world insights that can support better decision-making.

Research Question

The primary questions guiding this analysis are focused on both prediction accuracy and practical usability. First, we aim to determine how accurately PM10 levels can be forecasted for the following day using historical pollution and weather data. Second, we investigate which variables, including other pollutants and environmental conditions, contribute most significantly to predicting PM10 levels. Finally, we evaluate which modeling approach provides the best balance between predictive accuracy and interpretability, ensuring that the results are both reliable and understandable for practical use.

Data Description

The dataset used in this analysis was obtained from the Guanyuan air quality monitoring station in Beijing, which is part of a broader network of government-operated stations that continuously measure air pollution and environmental conditions. These monitoring systems are widely used in research and policy settings to track pollution trends and assess public health risks.

The dataset consists of hourly observations of PM10 levels, along with measurements of other pollutants such as nitrogen dioxide (NO₂), sulfur dioxide (SO₂), carbon monoxide (CO), and ozone (O₃). These pollutants are often associated with traffic emissions, industrial processes, and other sources of air contamination. In addition, the dataset includes weather-related variables such as temperature (TEMP), atmospheric pressure (PRES), dew point (DEWP), wind speed (WSPM), and rainfall (RAIN), all of which influence how pollutants accumulate or disperse in the atmosphere.

The data spans multiple years, providing a comprehensive view of both short-term fluctuations and long-term patterns in air quality. Because the data is recorded at an hourly frequency, it captures detailed changes in pollution levels throughout the day.

To prepare the data for modeling, several preprocessing steps were applied. Missing values were handled using interpolation methods to maintain continuity in the dataset, and outliers were identified using the interquartile range (IQR) method and replaced to reduce the influence of extreme values. The hourly data was then aggregated into daily averages, which aligns with the objective of forecasting next-day PM10 levels. Finally, lagged variables were created to incorporate information from previous days, reflecting the strong temporal dependence observed in pollution data.

Methodology

To forecast PM10 levels, a range of models with varying levels of complexity were developed and evaluated. The purpose of using multiple models is to understand how different approaches capture patterns in the data and to determine whether more complex methods provide meaningful improvements in prediction accuracy. By starting with simple models and gradually increasing complexity, we are able to clearly assess the value added by each additional component.

The naive model serves as a baseline and assumes that the PM10 level for the next day will be the same as the current day. Although this assumption is simplistic, it is useful because air pollution often exhibits persistence over time, meaning that today's conditions can be similar to tomorrow's. This model provides a benchmark for evaluating whether more advanced models are actually improving predictions beyond this basic assumption.

A simple linear regression model was then implemented using the previous day's PM10 value as the sole predictor. This model captures short-term temporal dependence by estimating a linear relationship between past and future pollution levels. In other words, it attempts to quantify how much of tomorrow's PM10 can be explained by today's value. While this improves

slightly over the naive model, it still ignores other important factors such as weather conditions and emissions from different sources.

To address these limitations, a multiple linear regression model was developed. This model incorporates lagged PM10 values along with additional predictors such as temperature, wind speed, rainfall, and other pollutants (NO₂, SO₂, CO). These variables are included because they represent real-world drivers of air pollution. For example, higher wind speeds can disperse pollutants, while emissions-related variables such as NO₂ and CO are often associated with traffic and industrial activity. By combining these predictors, the model is able to capture both temporal patterns and environmental influences, resulting in a more realistic and accurate representation of how PM10 levels are determined.

An ARIMA model was also constructed to capture time-series characteristics such as trends and autocorrelation. Unlike regression-based approaches, ARIMA models focus exclusively on patterns within the PM10 time series itself, such as repeating cycles or gradual changes over time. This makes ARIMA well-suited for modeling purely temporal relationships. However, because it does not incorporate external variables such as weather or other pollutants, its ability to fully explain variations in PM10 is limited, particularly when environmental factors play a significant role.

Finally, an XGBoost model was used as a machine learning approach to capture more complex relationships in the data. XGBoost builds a sequence of decision trees, where each new tree is designed to correct the errors made by previous ones. This process allows the model to learn nonlinear relationships and interactions between variables that may not be captured by linear models. For example, the combined effect of temperature and wind speed on pollution may not be strictly linear, and XGBoost can adapt to these types of patterns. Additionally, the model can automatically identify which variables are most important for prediction.

To evaluate model performance, three metrics were used: mean absolute error (MAE), root mean squared error (RMSE), and R². MAE measures the average difference between predicted and actual values, providing an intuitive measure of typical prediction error. RMSE places greater emphasis on larger errors, making it particularly useful for identifying models that struggle with extreme pollution levels. R² measures how much of the variation in PM10 levels is explained by the model, providing an overall indication of model fit. Together, these metrics offer a comprehensive view of both accuracy and reliability.

Results

The performance of the models is summarized in Table 1, which compares prediction accuracy across all approaches.

Model	MAE	RMSE	R²
Naive	44.58	60.01	0.32
Simple Linear	41.95	53.24	0.32
Multiple Linear	22.13	29.09	0.80
ARIMA	52.26	64.56	0.002
XGBoost	18.08	25.14	0.86

A clear progression in performance is observed as model complexity increases. The naive model, which assumes that tomorrow's PM10 level is equal to today's, produces relatively high prediction errors (MAE = 44.58, RMSE = 60.01). Despite its simplicity, the naive model still achieves an R² of approximately 0.32, indicating that PM10 levels exhibit a moderate degree of persistence over time. In other words, today's pollution level provides some information about tomorrow's level, but this information alone is not sufficient for accurate prediction.

The simple linear regression model, which uses the previous day's PM10 value as a predictor, produces only a small improvement in prediction accuracy compared to the naive model. While the MAE and RMSE decrease slightly, the R² remains essentially unchanged at 0.32. This suggests that modeling the relationship between consecutive days using a linear structure does not significantly improve predictive power beyond the naive assumption. This result highlights an important insight: although PM10 levels are temporally correlated, relying solely on past PM10 values does not capture the full set of factors influencing air pollution.

The multiple linear regression model shows a substantial improvement in performance, reducing the MAE from approximately 44 to 22 and the RMSE from 60 to 29. This represents nearly a 50% reduction in prediction error relative to the naive model. In addition, the R² increases significantly to 0.80, indicating that the model explains about 80% of the variability in PM10 levels. This improvement demonstrates the importance of incorporating additional predictors, such as weather conditions and other pollutants. Variables such as wind speed, rainfall, and emissions-related pollutants provide critical information about how pollution forms and disperses, allowing the model to capture more realistic patterns in the data.

In contrast, the ARIMA model performs poorly compared to the other approaches, with the highest prediction errors and an R^2 value close to zero. This indicates that the model explains almost none of the variation in PM10 levels. The weak performance of ARIMA suggests that relying solely on time-series structure, without incorporating external variables, is insufficient for modeling air pollution in this context. This result reinforces the idea that PM10 levels are strongly influenced by environmental and emission-related factors, rather than purely by historical trends.

The XGBoost model achieves the best overall performance, with the lowest MAE (18.08), lowest RMSE (25.14), and highest R^2 (0.86). This indicates that the model provides the most accurate predictions among all approaches considered. The improved performance of XGBoost can be attributed to its ability to capture nonlinear relationships and interactions between variables. For example, the combined effect of wind speed and pollutant levels on PM10 may not be strictly linear, and XGBoost is able to model these complex interactions more effectively than traditional regression methods.

However, it is important to note that the improvement of XGBoost over the multiple linear regression model is relatively modest. While XGBoost reduces prediction error further, the gain in performance is not dramatic. This suggests that much of the predictive signal in the data is already captured by the linear model, particularly due to the inclusion of lagged PM10 variables and key environmental predictors. In practical terms, this means that simpler models may still be sufficient in many cases, especially when interpretability is important.

Overall, these results indicate that incorporating additional predictors significantly improves model performance, while increasing model complexity beyond a certain point yields diminishing returns. The findings highlight the importance of balancing predictive accuracy with model simplicity, depending on the intended application.

From a practical perspective, these findings indicate that incorporating both environmental conditions and pollution-related variables leads to substantially more accurate predictions of PM10 levels. While simpler models provide a useful baseline, they fail to capture the complexity of real-world pollution dynamics. The improved performance of more advanced models demonstrates that air quality is influenced by multiple interacting factors, and accounting for these relationships is essential for producing reliable forecasts that can be used in real-world decision-making.

One limitation of the models is their reduced ability to accurately predict extreme pollution spikes, which occur less frequently and are inherently more difficult to capture due to their variability and dependence on sudden environmental or human-driven factors.

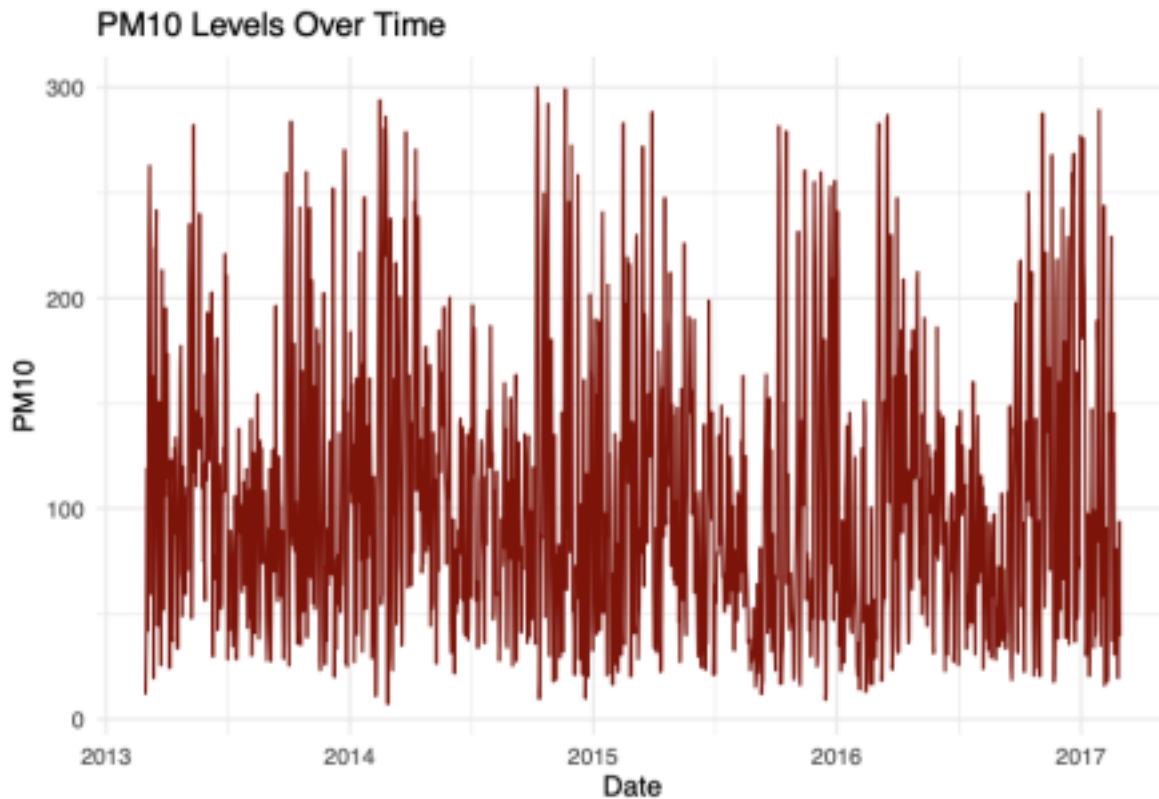
Classification Accuracy

In addition to predicting exact PM10 values, the models were also evaluated based on their ability to correctly classify pollution levels into meaningful categories, such as acceptable, unhealthy, and very unhealthy. This type of evaluation is particularly important for public-facing applications, where individuals are more concerned with understanding risk levels rather than precise numerical values. The models demonstrated strong performance in identifying general pollution categories, especially for typical pollution levels. However, similar to the results observed in numerical predictions, extreme pollution events remain more challenging to classify accurately. Despite this limitation, the models provide reliable guidance for most day-to-day conditions, making them useful tools for practical decision-making.

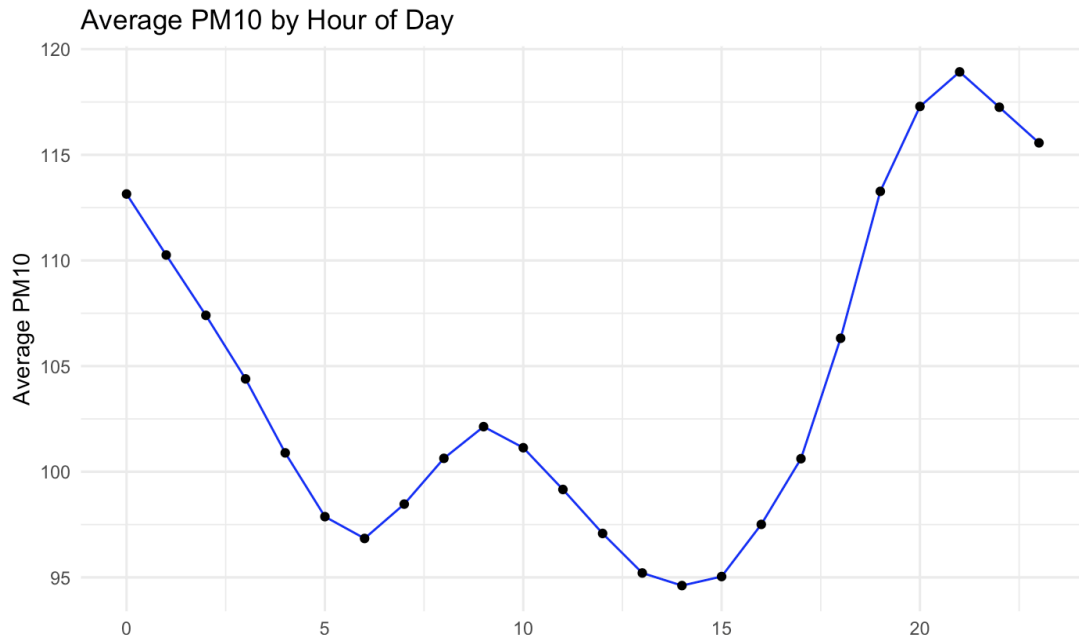
Exploratory Analysis and Visual Insights

To better understand the underlying patterns in PM10 levels and to support the development of predictive models, several exploratory analyses were conducted. These analyses focus on identifying temporal trends, seasonal patterns, relationships between variables, and how well the models capture observed pollution levels.

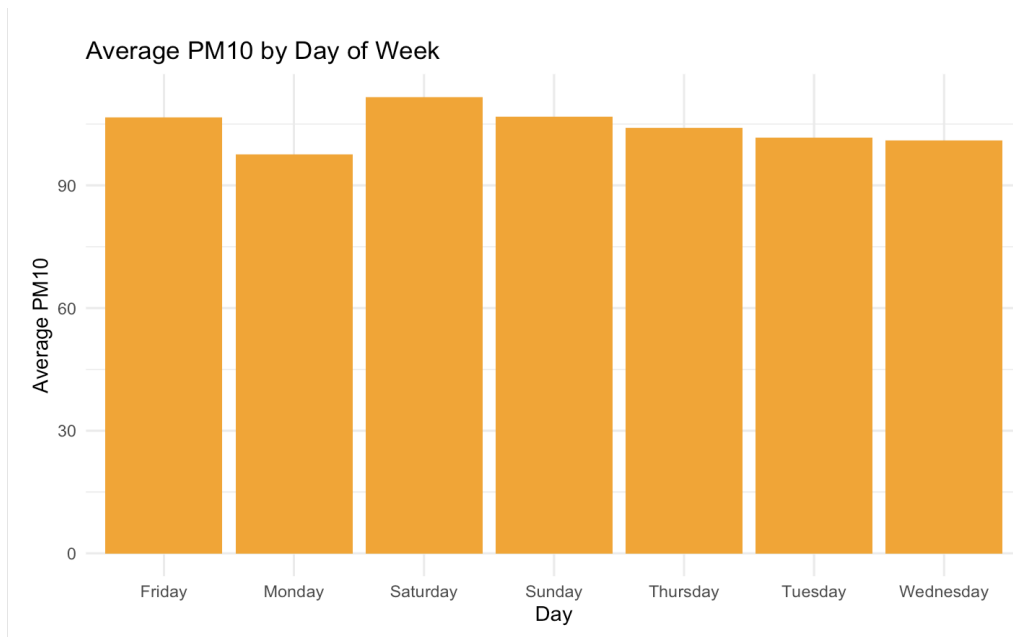
The time series plot of PM10 levels over the observed period reveals substantial variability, with frequent spikes indicating periods of elevated pollution. As shown in Figure 1, PM10 values fluctuate significantly from day to day, with some peaks exceeding 250–300. These spikes suggest that pollution levels can change rapidly due to a combination of environmental and human-related factors. In addition to short-term fluctuations, there are indications of longer-term patterns, although no consistent upward or downward trend is clearly observed over the entire time period. This variability highlights the importance of forecasting, as it is difficult to predict pollution levels based on visual inspection alone.



The hourly average plot shows a clear pattern in PM10 levels throughout the day. Concentrations tend to peak between 8 and 9 PM, while the lowest levels occur during the afternoon hours. This pattern suggests that time of day plays a significant role in air pollution levels. One possible explanation for the evening peak is increased human activity, such as traffic congestion during evening hours and emissions from residential energy use. In addition, atmospheric conditions later in the day, such as lower wind speeds or temperature inversions, can trap pollutants closer to the ground, allowing them to accumulate.

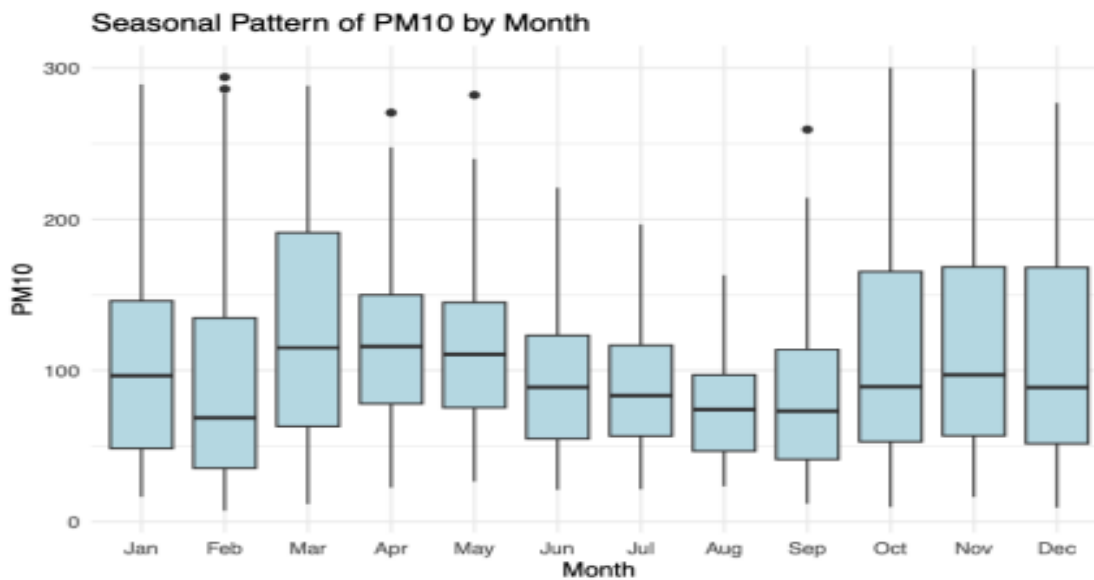


In terms of daily averages throughout the week, there is a constant increasing trend from Monday to the weekends, with Saturday showing the highest average PM10 levels. This pattern may reflect differences in human activity and traffic patterns across the week. Weekends potentially have more outdoor activities or different emission sources contributing to higher pollution levels. However, the differences in PM10 levels are relatively small.



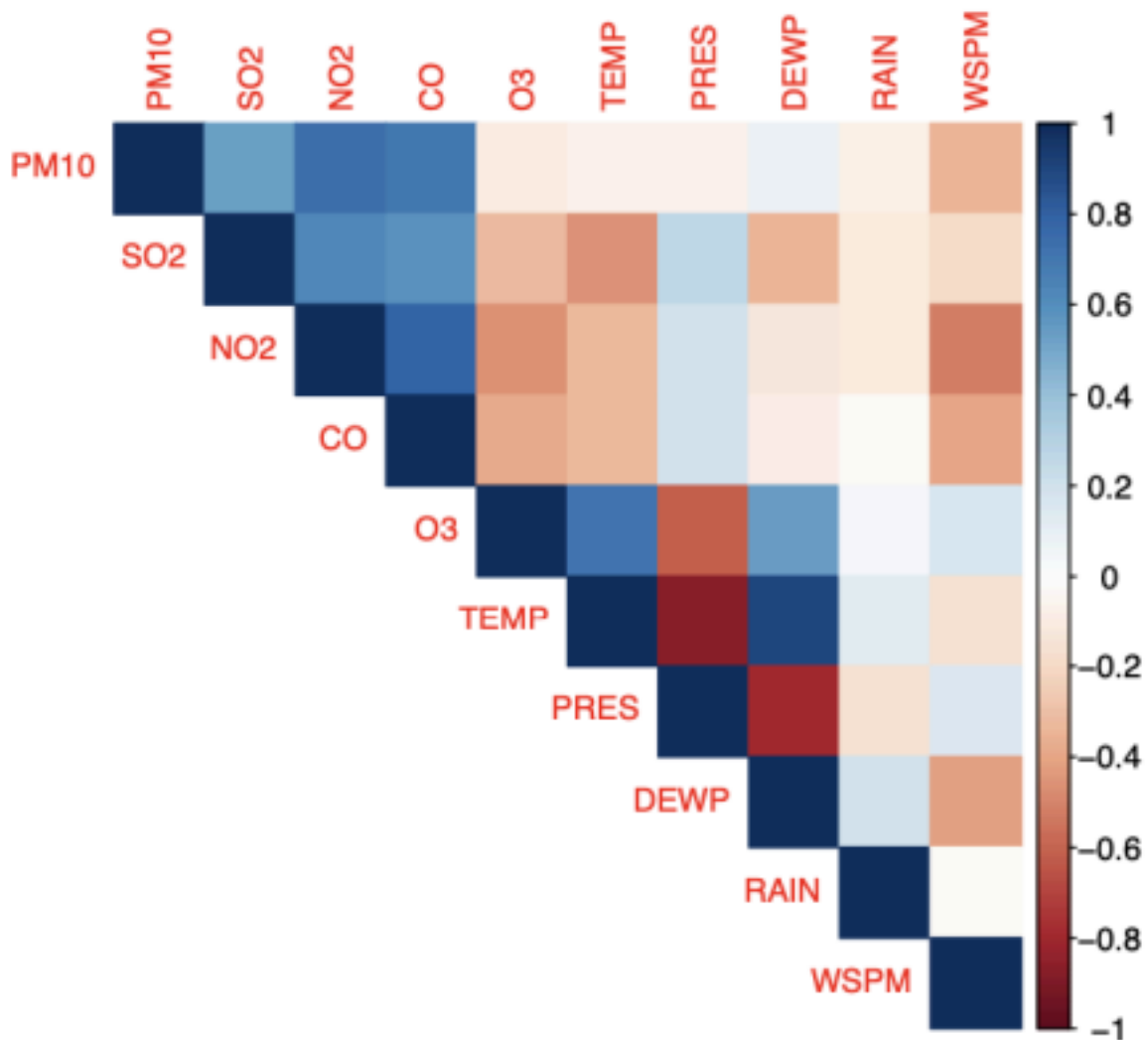
Further insights are obtained by examining seasonal patterns in PM10 levels. Figure 2 presents the distribution of PM10 values across different months using boxplots. The results indicate that pollution levels tend to vary throughout the year, with certain months exhibiting higher median values and greater variability. In particular, months such as March, October, and November appear to have higher pollution levels, while summer months such as July and August show relatively lower and more stable PM10 concentrations. This seasonal variation suggests that environmental conditions, such as temperature and atmospheric circulation, play a role in influencing pollution levels.

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To better understand the relationships between PM10 and other variables, a correlation heatmap is shown in Figure 3. The heatmap reveals strong positive correlations between PM10 and pollutants such as NO₂, SO₂, and CO. These relationships suggest that these pollutants may share common sources, such as traffic emissions or industrial activity. In contrast, wind speed (WSPM) shows a negative correlation with PM10, indicating that stronger winds help disperse pollutants and reduce their concentration. Other weather variables, such as temperature and pressure, exhibit weaker relationships with PM10, suggesting that their effects may be more complex or indirect.

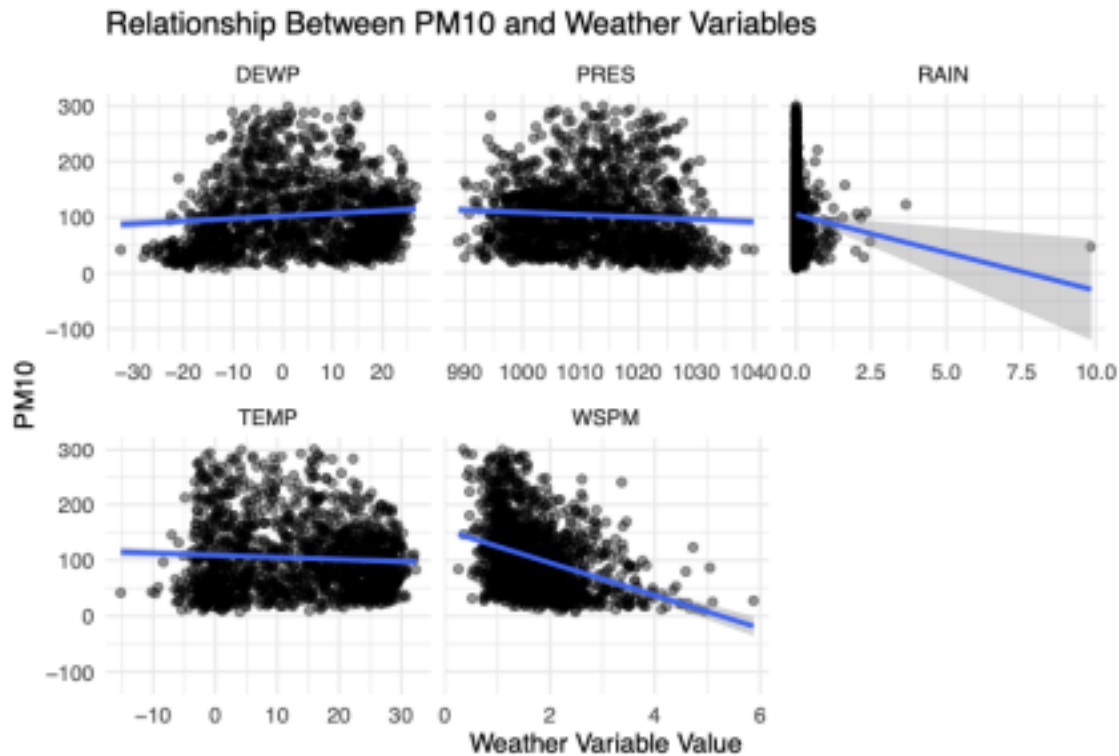
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Additional insight into these relationships is provided by the scatter plots shown in Figure 4, which illustrate the association between PM10 and selected weather variables. The plots indicate that wind speed has the most pronounced negative relationship with PM10, with higher wind speeds corresponding to lower pollution levels. Rainfall also appears to have a slight

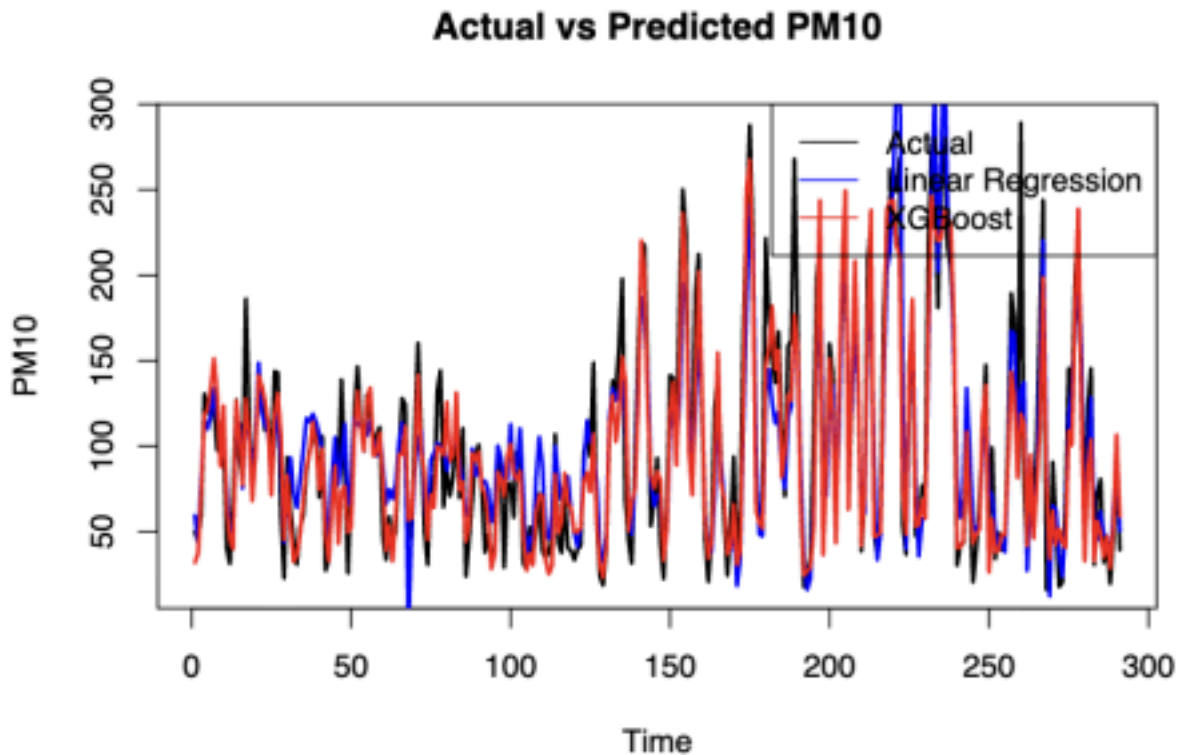
negative effect on PM10, although the relationship is less consistent. Other variables, such as temperature and dew point, show weaker and more dispersed patterns, indicating that their influence on PM10 is less direct or may depend on interactions with other factors.

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Finally, the performance of the predictive models is evaluated through a comparison of actual and predicted PM10 values, as shown in Figure 5. Both the multiple linear regression and XGBoost models are able to capture the general trends in the data, closely following the overall pattern of PM10 levels over time. However, the models struggle to accurately predict extreme spikes in pollution, often underestimating the highest values. This limitation is common in predictive modeling, as extreme events are less frequent and more difficult to capture. Despite this, the models perform well in predicting typical pollution levels, which are most relevant for everyday decision-making.

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Overall, these exploratory analyses provide important context for understanding PM10 dynamics and support the modeling approach used in this study. The observed relationships between pollution, weather, and time-based patterns reinforce the importance of incorporating multiple predictors in forecasting models and help explain the improvements observed in model performance.

Interpretation for Public Use

PM10 levels can be translated into meaningful categories that reflect overall air quality and the potential impact on human health. PM10 refers to fine particulate matter that can be inhaled into the lungs, making it an important indicator when assessing environmental safety. Because these particles are small enough to enter the respiratory system, higher concentrations are associated with increased health risks, particularly for vulnerable populations.

In general, PM10 levels below 70 are considered acceptable and pose minimal risk to most individuals. As levels rise to around 150, air quality becomes unhealthy, especially for sensitive groups such as children, the elderly, and individuals with pre-existing respiratory conditions. At levels above 225, air quality is considered very unhealthy, meaning that even the

general population may experience negative health effects, including difficulty breathing, irritation, and increased risk of illness.

These thresholds provide a useful framework for interpreting the predictions generated by the models. Rather than viewing predicted PM10 values as abstract numbers, they can be directly linked to real-world health implications. For example, a predicted PM10 level of 80 may indicate relatively safe conditions, while a prediction above 150 would signal a need for caution.

By combining model predictions with these categories, the results can be translated into actionable guidance. Individuals may choose to limit outdoor activities, wear protective masks, or take other precautions when high PM10 levels are expected. In addition, this information can be used by organizations and policymakers to issue air quality advisories or implement temporary measures to reduce exposure.

Overall, this interpretation step is critical because it connects the technical modeling results to real-world decision-making. It ensures that the predictions are not only accurate, but also meaningful and useful for improving public awareness and health outcomes.

Application of the Model

The developed models can be used to forecast PM10 levels for the next day by inputting current pollution and weather conditions into the trained model. This provides a practical and efficient tool for predicting air quality in advance, allowing users to anticipate potential changes rather than react after pollution levels have already increased. Because the model incorporates both historical pollution patterns and environmental variables, it is able to generate predictions that reflect real-world conditions more accurately than simple trend-based approaches.

For example, using recent data, the models predict PM10 levels in the range of approximately 70–80 for the following day. While this range falls within the “acceptable” category, it still reflects moderate pollution levels that may vary depending on local conditions. These predictions can then be interpreted using established PM10 thresholds to assess potential health risks and determine whether precautionary measures are necessary.

In practice, this type of forecasting can support a wide range of applications. Public health agencies can use the predictions to issue early warnings or advisories when elevated pollution levels are expected. Environmental organizations can use the model to monitor trends and evaluate the effectiveness of pollution control strategies. On an individual level, people can use these predictions to make informed daily decisions, such as adjusting outdoor activities, planning travel, or taking protective measures during high pollution periods.

Additionally, the model framework can be extended or updated as new data becomes available, allowing for continuous improvement in prediction accuracy. This flexibility makes the model not only useful for current forecasting but also adaptable for future applications in environmental monitoring and policy development.

Conclusion

This analysis demonstrates that PM10 levels can be effectively forecasted using a combination of historical pollution data and environmental variables. The results show that while simple models such as the naive and simple linear approaches provide a useful starting point, they are limited in their ability to capture the complexity of air pollution dynamics. Incorporating additional predictors, such as weather conditions and other pollutants, leads to substantial improvements in predictive accuracy.

The findings also highlight that PM10 levels are influenced by multiple interacting factors, including temporal patterns, environmental conditions, and relationships between different pollutants. The strong performance of the multiple linear regression model suggests that much of the underlying structure in the data can be captured using relatively simple and interpretable relationships. At the same time, the XGBoost model achieves the highest level of accuracy by capturing nonlinear effects and interactions that are not easily modeled using traditional approaches.

However, the relatively modest improvement of XGBoost over the multiple linear regression model suggests that increasing model complexity does not always lead to significantly better performance. This highlights an important trade-off between model accuracy and interpretability. In many practical applications, simpler models may be preferred because they are easier to understand and communicate, even if they are slightly less accurate.

One limitation of this analysis is the reduced ability of the models to accurately predict extreme PM10 spikes. These events are relatively rare and often driven by sudden changes in weather conditions or human activity, making them difficult to capture using standard predictive approaches. Additionally, the models do not explicitly account for external factors such as policy, industrial activity changes, which can have significant impacts on air quality but are not directly captured in the dataset.

Overall, the models developed in this project provide a practical and effective framework for next-day air quality forecasting. By combining strong predictive performance with clear interpretation, the results can be used to support informed decision-making at both the individual and organizational levels. More broadly, this analysis demonstrates how data-driven approaches can be used to better understand environmental systems and translate complex information into

Appendices

Appendix A: Data Processing and Preparation

Prior to modeling, several data preprocessing steps were performed to ensure the dataset was suitable for analysis and prediction. The original dataset consisted of hourly observations of pollution and weather variables collected from the Guanyuan monitoring station in Beijing. Because the objective of this project was to forecast next-day PM10 levels, the data was aggregated into daily averages to align with the prediction goal and simplify modeling.

Missing values were handled to maintain continuity in the dataset and avoid disruptions in time-based patterns. Extreme outliers were identified using the interquartile range (IQR) method and were replaced to reduce their influence on model performance while preserving the overall distribution of the data. This step was particularly important given the presence of occasional extreme pollution spikes that could disproportionately affect model training.

In addition, lagged variables were created to capture the temporal dependence in pollution levels. Specifically, previous-day PM10 values were included as predictors, as pollution levels tend to persist over time and are often influenced by prior conditions. These preprocessing steps ensured that the dataset was clean, consistent, and appropriately structured for forecasting next-day PM10 levels.

Appendix B: Model Outputs

This section provides additional technical details for the models discussed in the main report, including information that supports the results but is not necessary for the main narrative.

Linear and Multiple Regression Models

The regression models estimate the relationship between PM10 and a set of predictor variables, including lagged PM10 values, weather variables, and other pollutants. The model outputs include estimated coefficients, standard errors, and significance levels, which indicate the strength and reliability of each predictor.

The multiple linear regression model demonstrates that several predictors, particularly lagged PM10 values and pollutant variables such as NO₂ and CO, have strong associations with PM10 levels. These results are consistent with expectations, as these variables are closely tied to emission sources and environmental conditions that influence air pollution.

The ARIMA model was implemented to capture temporal patterns such as trends and autocorrelation in PM10 levels. Model outputs include parameter estimates and diagnostic measures used to evaluate model fit.

However, the results indicate that the ARIMA model performs poorly compared to other approaches. This is likely due to its reliance solely on past PM10 values, without incorporating external variables such as weather or other pollutants. As a result, the model is unable to fully capture the factors that influence changes in air quality.

XGBoost Model

The XGBoost model was trained using a combination of pollution variables, weather conditions, and lagged PM10 values. This model builds a sequence of decision trees that iteratively improve prediction accuracy by minimizing errors.

Feature importance analysis indicates that lagged PM10 values, along with pollutants such as NO₂ and CO, are among the most influential predictors. This confirms that both historical pollution levels and environmental conditions play a critical role in determining PM10 concentrations. The flexibility of the XGBoost model allows it to capture nonlinear relationships and interactions between variables, contributing to its strong predictive performance.

Appendix C: Additional Visualizations

This section includes additional diagnostic and supporting plots that provide further insight into the data and model performance. These visualizations were not included in the main report in order to maintain clarity and focus, but they offer useful technical detail for a deeper understanding of the analysis.

Examples of additional visualizations include residual plots for regression models, which help assess model fit and identify patterns in prediction errors, as well as feature importance plots from the XGBoost model. Additional scatter plots may also be included to further explore relationships between variables.

These visualizations support the conclusions presented in the main report and provide additional evidence for the modeling approach.

Appendix D: Code and Implementation

All analyses were conducted using the R programming language, including data cleaning, visualization, and model development. The full implementation is provided in the accompanying

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R Markdown file, which contains all code used to generate the results and visualizations presented in this report.

The code includes data preprocessing steps such as handling missing values, creating lagged variables, and aggregating data, as well as model training and evaluation procedures. Providing the full code ensures that the analysis is reproducible and allows others to replicate or extend the results if needed

Appendix E: Example Prediction

To demonstrate the practical application of the developed models, an example prediction is provided using recent observed data. This example illustrates how the model can be used to forecast PM10 levels for the following day based on current environmental conditions.

In this scenario, the current day's PM10 level is approximately 75, along with corresponding weather conditions such as temperature, wind speed, and rainfall. These values are used as inputs into the trained models to generate predictions for the next day.

Using the multiple linear regression model, the predicted PM10 level for the following day is approximately 71. Similarly, the XGBoost model produces a prediction of approximately 79. Both predictions fall within a similar range, indicating consistency between the models and suggesting a moderate level of air pollution for the next day.

These predicted values can then be interpreted using the PM10 thresholds discussed in the main report. A predicted level in the range of 70–80 suggests that air quality is generally acceptable but may still warrant attention for sensitive individuals. This type of prediction allows users to anticipate air quality conditions and make informed decisions accordingly.

This example highlights the usability of the developed models as a forecasting tool. By inputting current pollution and weather data, the models provide a straightforward method for estimating next-day PM10 levels. This demonstrates how the analysis can be translated into a practical, real-world application that supports both individual decision-making and broader environmental awareness.